

Problem 6

A target consists of ten concentric circles with radii r_k ($k = 1, 2, \dots, 10$), where $r_1 < r_2 < \dots < r_{10}$. Event A_k denotes a hit in circle r_k ($k = 1, 2, \dots, 10$). What do the events

$$B = \sum_{k=1}^6 A_k \quad \text{and} \quad C = \prod_{k=1}^{10} A_k$$

mean?

Solution

We need to interpret these events in the context of the concentric circles target.

Setup and Notation

Target Description

The target consists of 10 concentric circles with increasing radii:

$$r_1 < r_2 < r_3 < \dots < r_{10}$$

The innermost circle has radius r_1 (typically the bullseye), and the outermost circle has radius r_{10} .

Event Definition

A_k = "hit lands within the circle of radius r_k "

This means: $A_k = \{(x, y) : x^2 + y^2 \leq r_k^2\}$

Important Observation: Nested Structure

Since $r_1 < r_2 < \dots < r_{10}$, the circles are nested:

$$A_1 \subset A_2 \subset A_3 \subset \dots \subset A_{10}$$

This means:

- Any hit in A_1 is automatically in $A_2, A_3, \dots, A_{10}$
- Any hit in A_k is automatically in $A_{k+1}, A_{k+2}, \dots, A_{10}$

This nested structure is crucial for interpreting the events B and C .

Part 1: Event $B = \sum_{k=1}^6 A_k$

Definition

$$B = \sum_{k=1}^6 A_k = A_1 + A_2 + A_3 + A_4 + A_5 + A_6 = \bigcup_{k=1}^6 A_k$$

This is the union of the first six events.

Interpretation Using Nested Structure

Since $A_1 \subset A_2 \subset \dots \subset A_6$, the union of all these sets is simply the largest one:

$$A_1 \cup A_2 \cup \dots \cup A_6 = A_6$$

This is because any point in $A_1, A_2, \dots, A_5$ is already contained in A_6 .

Conclusion for Event B

$$\boxed{B = A_6}$$

Meaning: Event B represents a hit within the circle of radius r_6 (the 6th circle from the center).

In other words: The hit lands within the inner six circles, or equivalently, within distance r_6 from the center.

Part 2: Event $C = \prod_{k=1}^{10} A_k$

Definition

$$C = \prod_{k=1}^{10} A_k = A_1 \cdot A_2 \cdot A_3 \cdots A_{10} = \bigcap_{k=1}^{10} A_k$$

This is the intersection of all ten events.

Interpretation Using Nested Structure

Since $A_1 \subset A_2 \subset \cdots \subset A_{10}$, the intersection of all these sets is simply the smallest one:

$$A_1 \cap A_2 \cap \cdots \cap A_{10} = A_1$$

This is because A_1 is contained in all other sets, so the common part of all sets is just A_1 .

Conclusion for Event C

$$\boxed{C = A_1}$$

Meaning: Event C represents a hit within the innermost circle of radius r_1 (the bullseye).

In other words: The hit lands in all ten circles simultaneously, which means it must be in the smallest circle.

General Principle for Nested Sets

For any nested sequence of sets $S_1 \subset S_2 \subset \cdots \subset S_n$:

- $\bigcup_{k=1}^n S_k = S_n$ (union equals the largest set)
- $\bigcap_{k=1}^n S_k = S_1$ (intersection equals the smallest set)

Visualization

Concentric Circles Target

Imagine concentric circles where:

- Circle 1 (innermost, bullseye): radius r_1
- Circle 2: radius r_2
- $\vdots$
- Circle 6: radius r_6
- $\vdots$
- Circle 10 (outermost): radius r_{10}

Then:

- Event B : Hit anywhere within circle 6 (including circles 1-5)
- Event C : Hit in the bullseye (circle 1)

Alternative Interpretation

If we instead define A_k as "hit lands in the annular region between circles $k - 1$ and k " (i.e., the ring), then the events would be disjoint, and:

- $B = A_1 \cup A_2 \cup \dots \cup A_6$ would mean "hit in any of the first 6 rings"
- $C = A_1 \cap A_2 \cap \dots \cap A_{10} = \emptyset$ would be impossible (empty set)

However, based on the problem statement that A_k denotes "hit in circle r_k " (not "ring k "), the nested interpretation is correct.

Final Answer

Given that $A_k =$ "hit within circle of radius r_k " and $A_1 \subset A_2 \subset \dots \subset A_{10}$:

Event B :

$$B = \sum_{k=1}^6 A_k = A_6$$

Meaning: A hit within the 6th circle from the center (within distance r_6 from the center). This includes hits in any of the inner six circles.

Event C :

$$C = \prod_{k=1}^{10} A_k = A_1$$

Meaning: A hit in the innermost circle (bullseye), within distance r_1 from the center. This is the only region that belongs to all ten circles simultaneously.

Key Insight: Due to the nested structure of concentric circles:

- The union of nested sets equals the largest set
- The intersection of nested sets equals the smallest set